Tasks for Prospective GSoC 2026 Applicants for the
PrediCT Projects

Submission Guidelines

- **IN ORDER TO BE CONSIDERED AS AN APPLICANT FOR GOOGLE SUMMER OF CODE YOU MUST ALSO SUBMIT A PROPOSAL THROUGH THE GOOGLE SUMMER OF CODE PORTAL**
- Additional tasks are listed below
- When completed, please use this Google form to submit

Resources:

Follow the instructions in this github to download the Stanford COCA dataset:
KatyEB/PrediCT at GSoC

Below are the tasks that we will use to evaluate prospective students for **PrediCT.**
After completing the first common test, please thoroughly complete the specific second test for your project of interest and (optionally) other second tests if you would like to also be considered for additional PrediCT projects at the same time (this may increase your chances of success, but make sure you don't do it at the expense of the specific project you are interested in).

Note: please work in your own github branch (i.e. NO PRs should be made). Send us a link to your code when you are finished and we will evaluate it.

Common Task (Required for All Projects): COCA Dataset Preprocessing

Goal: Build a preprocessing and data loading pipeline tailored to your target project

Tasks:
1. Follow PrediCT GitHub instructions to download and resample the COCA dataset
2. Implement: (a) HU windowing for cardiac CT, (b) Data augmentation suitable for your task, (c) Stratified train/val/test split, (d) Sampling strategy for class imbalance (if relevant)
3. Create an efficient data loader
4. Tailor to your project**:** Project 1 - optimize for segmentation; Project 2 - ensure radiomics compatibility; Project 3 - prepare for template extraction

Deliverables: Pipeline code, data loader, written justification (1-2 paragraphs), dataset statistics

Resources:

3D Slicer: Download 3D Slicer | 3D Slicer

MONAI: Project-MONAI/MONAI: AI Toolkit for Healthcare Imaging

Pydicom: pydicom documentation — pydicom 3.0.1 documentation

SimpleITK: SimpleITK - Home

Specific Task: Project 1 (Calcium Segmentation) - Heart Segmentation Model

Goal: Demonstrate medical image segmentation and model evaluation skills

Tasks:
1. Obtain free TotalSegmentator license, use it to segment whole-heart volumes in 30-50 COCA scans (establishes ground truth)
2. Train a lightweight model (U-Net or choice) to predict heart bounding box/coarse mask, preferably faster than TotalSegmentator
3. Evaluate: Dice score vs. TotalSegmentator on test set (target: >0.85), compare inference times

Deliverables: Model weights, evaluation notebook with Dice scores/times/visualizations, justification (1 paragraph) for model choice

Resources:

TotalSegmentator: wasserth/TotalSegmentator: Tool for robust segmentation of >100 important anatomical structures in CT and MR images

TotalSegmentator license: TotalSegmentator License

Specific Task: Project 2 (Radiomics & Phenotyping) - Feature Extraction

Goal: Demonstrate radiomics feature extraction and statistical analysis

Tasks:
1. Extract features using PyRadiomics from 20-30 COCA scans:
    a. Shape: Sphericity, SurfaceVolumeRatio, Maximum3DDiameter
    b. GLCM: Contrast, Correlation, InverseDifferenceMoment
    c. GLSZM: SmallAreaEmphasis, LargeAreaEmphasis, ZonePercentage
    d. GLRLM: ShortRunEmphasis, LongRunEmphasis, RunPercentage
    e. Optional: Max/mean calcium HU, total volume
2. Calculate Agatston scores (use original spacing), categorize: 0, 1-99, 100-399, ≥400

3. Statistical analysis: Correlate features with Agatston scores (Spearman, Kruskal-Wallis), report p-values and coefficients
4. Optional: t-SNE/UMAP visualization, unsupervised clustering, phenotype characterization

Deliverables: Extraction code, CSV with features, analysis notebook (correlation matrix, significant associations $p<0.05$, visualizations), optional t-SNE plot

Resources:

PyRadiomics: [Welcome to pyradiomics documentation! — pyradiomics v3.1.0rc2.post5+g6a761c4 documentation]

Specific Task: Project 3 (Physics-Based Simulation) - Coronary Atlas Registration

Goal: Demonstrate medical image registration and validation

Tasks:

1. Download ImageCAS CCTA coronary atlas from Kaggle (includes centerlines/vessels)
2. Register atlas to 20-30 non-contrast COCA scans using tool of choice (Elastix, ANTs, SimpleITK, custom)
    a. Challenge: CCTA (contrast) → NCCT (non-contrast) has different tissue contrasts
3. Balance accuracy vs. speed: Report registration time/scan, justify strategy (multi-resolution? rigid vs. deformable? metric?)
4. Validate: Calculate % of calcium voxels inside transformed vessel zones (target: >70% within ±10mm of centerlines), show visual examples

Deliverables: Registration code with parameters, validation notebook (accuracy metrics, timing, visual overlays), justification (1-2 paragraphs) addressing accuracy/speed tradeoff and challenges

Expected Challenges: CCTA-to-NCCT contrast differences, patient positioning variability, cardiac motion artifacts

Resources:
ImageCAS (Kaggle): ImageCAS | Kaggle

Elastix: elastix

ANTs: ANTsX/ANTs: Advanced Normalization Tools (ANTs)

SimpleITK registration tutorials:
https://simpleitk.readthedocs.io/en/v2.5.0/registrationOverview.html#registration-overview

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